Home Safety System

Wearable Fall Detection & Smart Stove Control

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Abstract

This wearable safety system protects elderly individuals living alone by combining automatic fall detection with an emergency-response workflow and an automated stove-safety cutoff.

A pocket-worn unit reads a Memsic 2125 dual-axis accelerometer via pulse-width sampling on a BASIC Stamp 2 microcontroller and detects falls as calibrated deviations from a per-session baseline. On detection it opens a cancel window; if not cancelled, it dispatches an emergency alert with location and, in parallel, cuts power to the stove.

Technical Highlights

- **Reliable fall detection:** a Memsic 2125 accelerometer sampled by pulse width, with per-session baseline calibration for accurate deviation detection.
- **Grace period:** a 30-second cancel window (20 s silent + 10 s buzzer) prevents false alarms before any alert is sent.
- **Location-aware alerting:** on a confirmed fall the unit commands the user's phone over an RN-42 Bluetooth link to text GPS coordinates and a street address to an emergency contact.
- **Smart stove safety:** a second BASIC Stamp 2, wired into a 15 A electric stovetop through a Solid State Relay, cuts stove power automatically.
- **User-friendly hardware:** soft-reset button, hardware kill switch, and status LEDs; a bill-of-materials analysis showed strong cost reduction at production scale.

Technologies Used

- **Language:** PBASIC
- **Controllers:** Dual BASIC Stamp 2 (master/slave)
- **Sensors:** Memsic 2125 dual-axis accelerometer (PWM sampling), GPS module
- **Communication:** RN-42 Bluetooth link to a mobile phone
- **Power control:** Solid State Relay switching a 15 A stovetop
- **Interface:** Buzzer, status LEDs, soft-reset and hardware kill switch

Outcome

The system delivers an end-to-end safety loop for independent living: it detects a fall, offers a cancel window, alerts a contact with precise location, and independently removes a common household fire hazard by cutting stove power — a compact, low-cost mechatronic safeguard.